

Yuli Multi-Enhancing Artificial Vision by Integrating Machine Learning in Retinal Prosthetics

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invertible Neural Networks

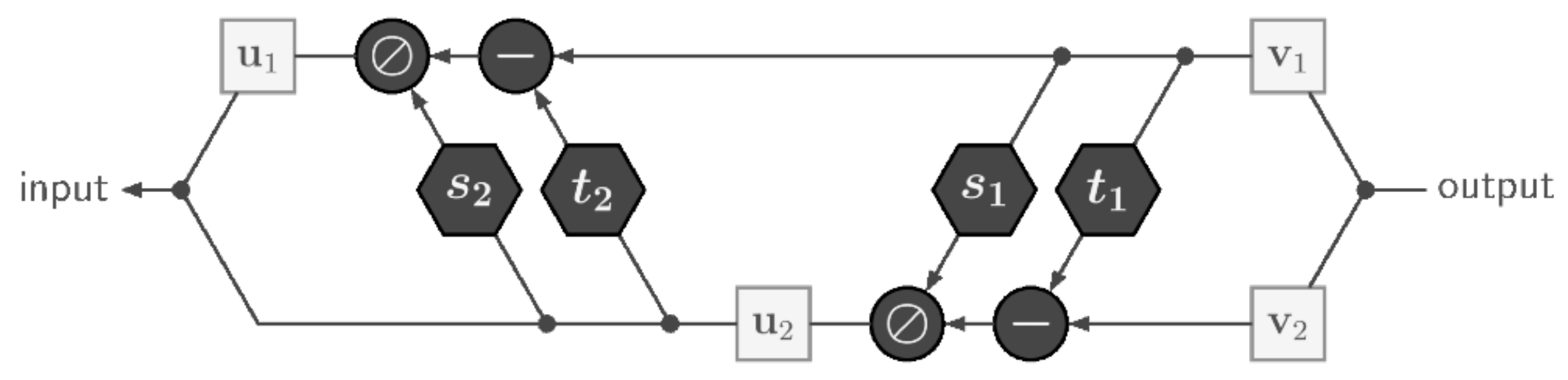
[Dinh et al., 2016]

$$\frac{\partial y}{\partial x^T} = \begin{bmatrix} \mathbb{I}_d & 0 \\ \frac{\partial y_{d+1:D}}{\partial x_{1:d}^T} & \text{diag} \left(\exp \left[s \left(x_{1:d} \right) \right] \right) \end{bmatrix} \qquad | \det \mathbf{J} | = \exp \left[\sum_j s \left(x_{1:d} \right)_j \right]$$

Fast Jacobian Determinant



<https://hci.uni-heidelberg.de/vision/inverse-problems-invertible-neural-networks/>

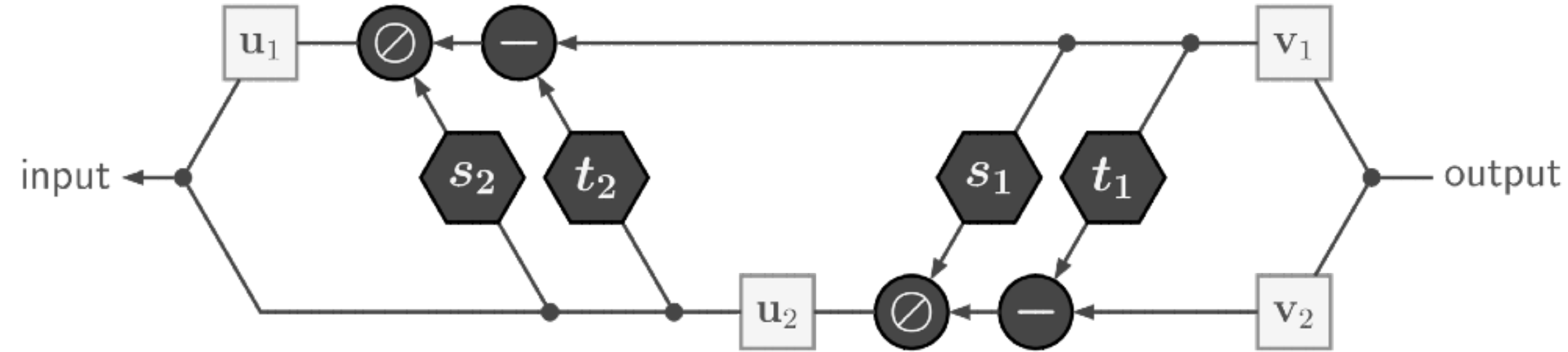


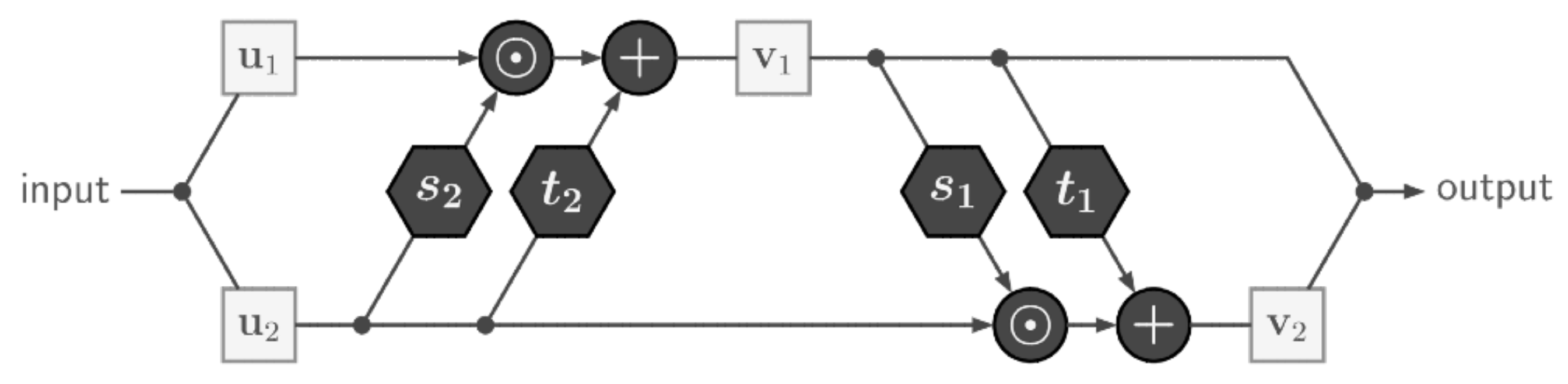
$$\mathbf{z} \sim \pi(\mathbf{z}), \mathbf{x} = f(\mathbf{z}), \mathbf{z} = f^{-1}(\mathbf{x})$$

$$p(\mathbf{x}) = \pi(\mathbf{z}) \left| \det \frac{d\mathbf{z}}{d\mathbf{x}} \right| = \pi(f^{-1}(\mathbf{x})) \left| \det \frac{df^{-1}}{d\mathbf{x}} \right|$$

Inventability







$$\mathbf{v}_1 = \mathbf{u}_1 \odot \exp\left(s_2(\mathbf{u}_2)\right) + t_2(\mathbf{u}_2)$$

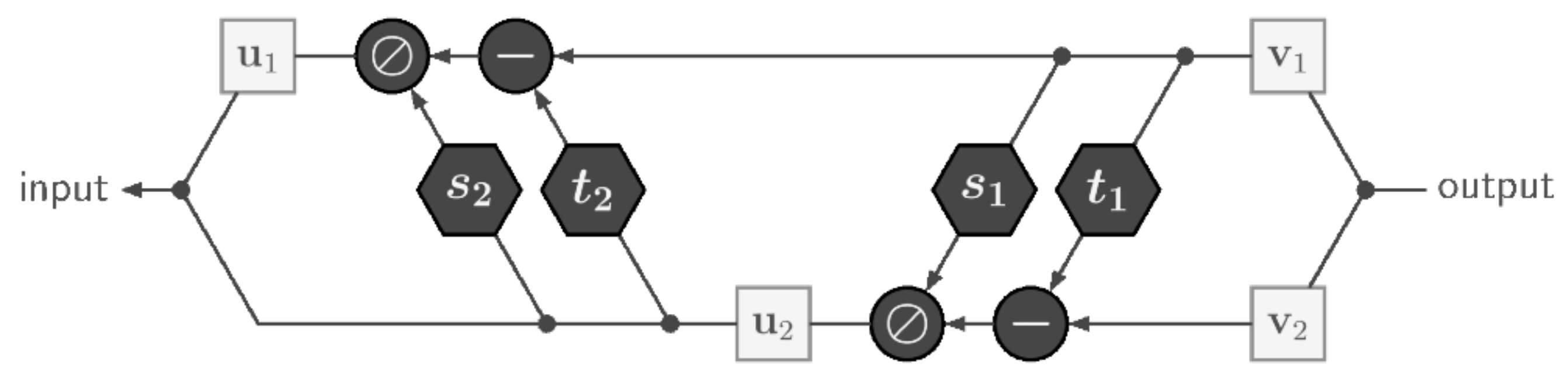
$$\mathbf{v}_2 = \mathbf{u}_2 \odot \exp\left(s_1(\mathbf{v}_1)\right) + t_1(\mathbf{v}_1)$$

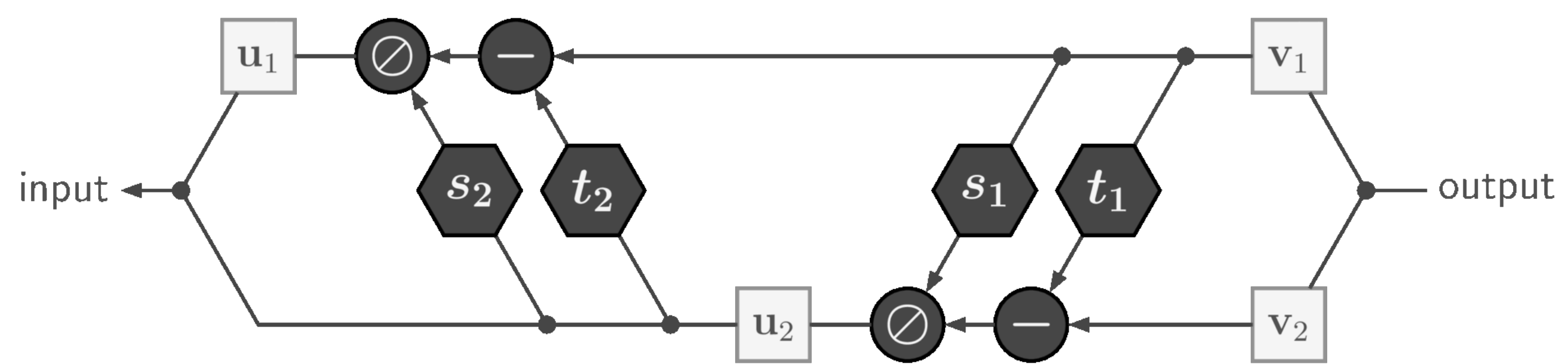
$$\mathbf{u}_2 = (\mathbf{v}_2 - t_1(\mathbf{v}_1)) \odot \exp(-s_1(\mathbf{v}_1))$$

$$\mathbf{u}_1 = (\mathbf{v}_1 - t_2(\mathbf{u}_2)) \odot \exp(-s_2(\mathbf{u}_2))$$





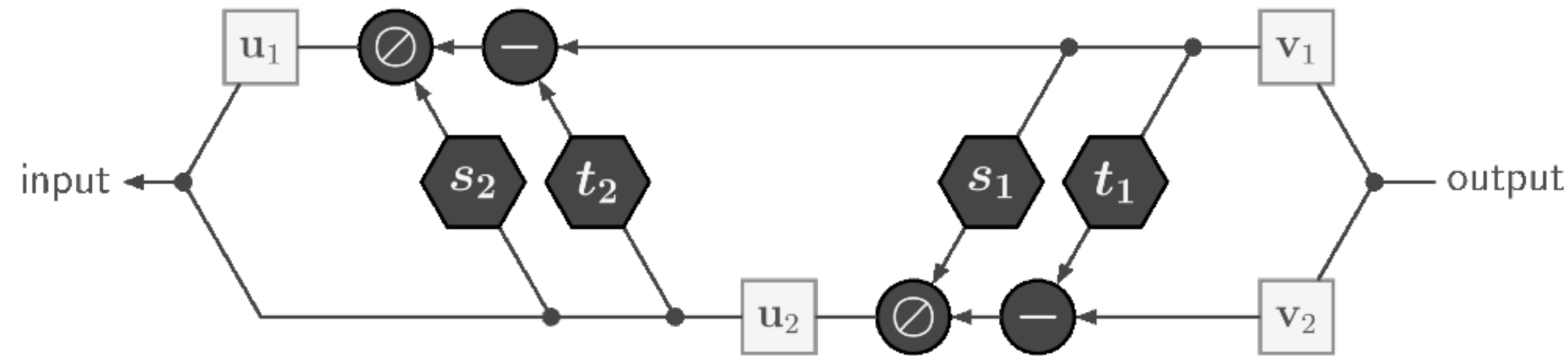




Invertible Neural Networks

$$\mathbf{z} \sim \pi(\mathbf{z}), \mathbf{x} = f(\mathbf{z}), \mathbf{z} = f^{-1}(\mathbf{x})$$

$$p(\mathbf{x}) = \pi(\mathbf{z}) \left| \det \frac{d\mathbf{z}}{d\mathbf{x}} \right| = \pi(f^{-1}(\mathbf{x})) \left| \det \frac{df^{-1}}{d\mathbf{x}} \right|$$



<https://hci.iwr.uni-heidelberg.de/vislearn/inverse-problems-invertible-neural-networks/>

$$\mathbf{v}_1 = \mathbf{u}_1 \odot \exp(s_2(\mathbf{u}_2)) + t_2(\mathbf{u}_2)$$

$$\mathbf{v}_2 = \mathbf{u}_2 \odot \exp(s_1(\mathbf{v}_1)) + t_1(\mathbf{v}_1)$$

$$\mathbf{u}_2 = (\mathbf{v}_2 - t_1(\mathbf{v}_1)) \odot \exp(-s_1(\mathbf{v}_1))$$

$$\mathbf{u}_1 = (\mathbf{v}_1 - t_2(\mathbf{u}_2)) \odot \exp(-s_2(\mathbf{u}_2))$$

$$\frac{\partial y}{\partial x^T} = \begin{bmatrix} \mathbb{I}_d & 0 \\ \frac{\partial y_{d+1:D}}{\partial x_{1:d}^T} & \text{diag}(\exp[s(x_{1:d})]) \end{bmatrix}$$

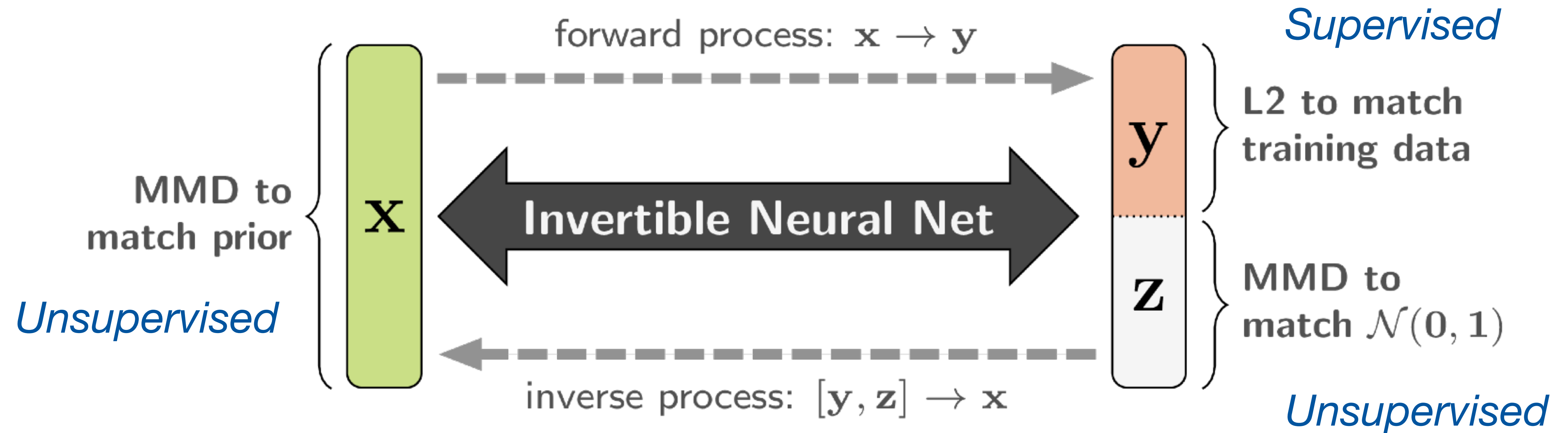
$$|\det \mathbf{J}| = \exp \left[\sum_j s(x_{1:d})_j \right]$$

Invertibility ✓

Fast Jacobian Determinant ✓

[Dinh et al., 2016]

MMD INN for Inverse Problems



$$\mathcal{L}_{\text{MMD}}(p_M, p_{M'}) = \left(\mathbb{E}_{i,j} [k(\mu_i, \mu_j) - 2k(\mu_i, \mu'_j) + k(\mu'_i, \mu'_j)] \right)^{\frac{1}{2}}$$

Maximum Mean Discrepancy [Gretton et al., 2006]